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TEMPERATURE AND WATER RELATIONS IN TWO SPECIES OF SPINY MICE (*ACOMYS*)

AMIRAM SHKOLNIK AND ARIEH BORUT

ABSTRACT.—Temperature regulation and water economy were studied in two species of spiny mice, *Acomys russatus* and *Acomys cahirinus*. Both species share the same desert habitat in Israel, but *A. russatus* is active during the day and *A. cahirinus* is nocturnal. It was shown that *A. russatus* was better able to maintain its normal body temperature (T_B) when exposed to high ambient temperatures (T_A) and survived temperatures as high as 42.5°C. On the other hand, T_A 40°C was lethal to *A. cahirinus*. Mean minimal metabolism of *A. russatus* was 34.5 per cent lower than the value predicted from body weight, and that of *A. cahirinus* was 13.5 per cent lower. *A. russatus* had no thermoneutral zone, oxygen consumption being minimal at T_A 30°C. *A. cahirinus* had a narrow thermoneutral zone between 27 and 32.5°C T_A . High ratios of evaporative water loss per milliliter of oxygen consumed were measured in both species; at 30°C it was 2.3 mg H_2O/ml O_2 in *A. russatus* and 1.9 mg H_2O/ml O_2 in *A. cahirinus*. When restrained at 30°C, 60 to 70 per cent of evaporation was from the skin. Maximum urine urea concentration in both species was 4700 to 4800 mM, among the highest recorded for mammals. Maximum chloride concentration was higher in the urine of *A. russatus* and amounted to 1500 mN, almost twice that of *A. cahirinus*. Accordingly, the capacity to utilize salt solutions for drinking was higher in *A. russatus*, which could drink sea water or NaCl solution as concentrated as 0.9 N. At 30°C, both species, when deprived of drinking water, failed to survive on a diet of barley. *A. russatus* could be considered physiologically better adapted for a diurnal existence than *A. cahirinus*, both in its ability to cope with high ambient temperatures and in its capacity to derive water from salty, succulent plants.

The common spiny mouse (*Acomys cahirinus*) and the golden spiny mouse (*Acomys russatus*) both are widely distributed and abundant in extreme deserts in Israel. They share the same rocky habitat and are similar in morphology and life history. They differ, however, in their daily activity patterns in that the common spiny mouse is a typically nocturnal rodent whereas the golden spiny mouse is frequently active during the daytime (Shkolnik, 1966). The high ambient temperatures and intense radiation encountered during the day could pose serious problems for temperature regulation and water balance. It is interesting, therefore, to find out whether the diurnal spiny mouse is better able to regulate its temperature in hot environments, and to derive more water from its food than its nocturnal relative.

Information on physiological adaptation to desert life in rodents is considerable. Most of the data have been obtained from desert species of such groups as Heteromyidae, Sciuridae, Cricetinae, and Dipodidae (see Schmidt-Nielsen, 1964), which comprise the predominant element of rodent fauna in the Nearctic and Palearctic deserts. A study of the physiology of the murid genus *Acomys*, which inhabits desert areas in southwest Asia and Africa, may disclose an additional pattern of adaptation to desert conditions.

MATERIALS AND METHODS

Individuals of the two species of spiny mice were trapped alive in the hot, dry deserts of southeastern Israel. Animals designated for physiological studies were kept in the laboratory for 1 to 3 months before being tested. During this period a mixed diet of seeds, vegetables, and land snails was offered. No free water was available. The mice easily adapted to captivity and seemed healthy. Breeding was frequent when both sexes were kept together. Only adult males, in good physical condition, were used in our experiments in order to avoid possible complications due to oestrous and pregnancy cycles. The average weight of *A. cahirinus* was 42.0 grams (71 individuals); for *A. russatus*, 51.1 grams (83 individuals).

Measurements of body temperature and oxygen consumption were made over a range of ambient temperatures between 5 and 45°C. Evaporation was measured between 20 and 32.5°C. At temperatures above 32.5°C, animals salivated excessively and saliva dripped from their mouths. Animals were acclimated for a week at temperatures within 5°C of the experimental temperatures before these measurements were made.

Body temperature was measured by a 35-gauge chromel-alumel thermocouple inserted 3 centimeters into the rectum and read on a Doran thermocouple potentiometer sensitive to 0.1°C. Measurements of body temperature (T_B) at different ambient temperatures (T_A) were taken after an animal had been kept under experimental thermal conditions for at least 4 hours.

Oxygen consumption was studied in an open circuit system (Depocas and Hart, 1957). The air was dried through columns of Drierite and passed at a rate of 280 milliliters per minute through a cylindrical glass jar where the animal was kept above a layer of paraffin oil. The jar was submerged in a regulated water bath. Temperature fluctuations in the jar did not exceed 0.5°C. At ambient temperatures below 20°C the air flow was raised to 350 milliliters per minute in order to avoid an increase of CO₂ concentration and decrease of O₂ beyond the limit of 1 per cent.

Determinations of oxygen concentration were made with a Beckman Model E paramagnetic oxygen analyzer. After an animal had equilibrated in the jar and oxygen consumption had become relatively constant, readings were taken at intervals of 5 minutes for a period of 80 minutes and the average was calculated. Readings taken while the animal was active were discarded. All results were corrected to standard temperature and pressure. The air flow maintained the relative humidity in the jar between 15 and 30 per cent. Animals used in these experiments were deprived of all food for 16 to 20 hours prior to the experiment in order to achieve a post-absorptive state.

Evaporative water loss was measured simultaneously with oxygen consumption. Water vapor was collected in glass U-tubes submerged in a freezing bath according to the method of Schmidt-Nielsen and Schmidt-Nielsen (1950). Four measurements of 20 minutes each were carried out after the system had equilibrated for an hour. Before each experiment, blank measurements were made under conditions identical to those of the experiment. The blank determination never exceeded 5 per cent of the amount of water vapor measured in the experiments. This value was subtracted from the amount of water measured with the animal in the system. Results of measurements taken when the animal was active were discarded.

An attempt was made to partition total evaporation into respiratory and cutaneous components. A mouse was placed in a plastic bag that covered its entire body surface with the exception of the snout, and then evaporation and oxygen consumption were measured. Cutaneous evaporation was trapped within the bag and only pulmonary evaporation was collected in the drying tubes. Restraint in the plastic bag, however, caused an elevation in the oxygen consumption of the mouse. Similar conditions of restraint, with a similar elevation in oxygen consumption, were obtained by enclosing the mouse in a net bag, which, presumably, had no effect on cutaneous evaporation. Evaporation at the higher levels

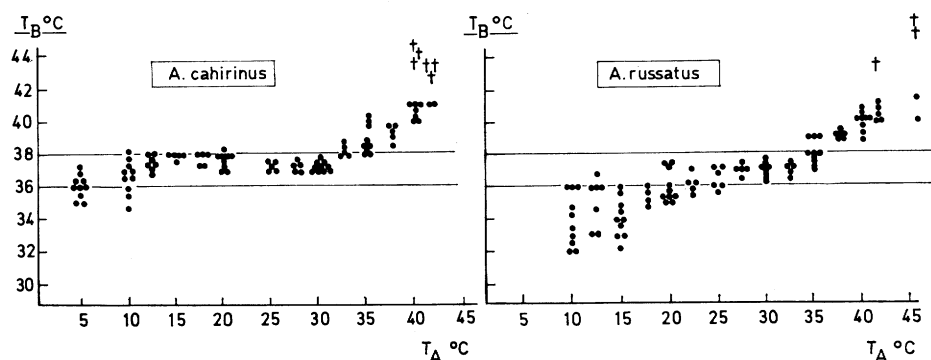


FIG. 1.—Body temperature (T_B) in spiny mice at different ambient temperatures (T_A). Each point represents one mouse. Crosses represent animals that died during exposure.

of oxygen consumption, therefore, could be partitioned by subtracting pulmonary evaporation (measured with the mouse in the plastic bag) from total evaporation (measured with the animal in the net bag). White mice (*Mus musculus*) were included in these experiments for purposes of comparison.

Ability to maintain body weight without drinking water was determined while keeping the animals on a diet of dry barley under three combinations of temperature and humidity: a) 30°C, 30 per cent RH (10 mg H_2O /l air); b) 26°C, 60 per cent RH (15 mg H_2O /l air); c) room conditions in summer 18 to 26°C, 67–75 per cent RH (11 to 18 mg H_2O /l air).

Ability to utilize NaCl solutions for drinking was determined with the animals kept individually in small metal cages. Dry soybeans were the only food available. Because of their high protein content, soybeans cause an increase in urea formation and urinary excretion, and result in an obligatory need for drinking water (Schmidt-Nielsen and Schmidt-Nielsen, 1952). NaCl solutions and sea water from the Mediterranean Coast (salinity 3.5 per cent) were offered in L-shaped drinking tubes. NaCl concentration in the drinking water was progressively increased in order to find the maximum concentration tolerable.

Capacity of the kidney to concentrate electrolytes in the urine was determined during the above-mentioned experiments. Excited animals were often found to have low urine concentrations. In order to minimize disturbances while urine samples were collected, the animal was retained in its cage, which was carefully turned upside down (in order to eliminate contamination of the samples) and placed over parafilm. Samples of urine were collected immediately when voided and sealed in capillary glass tubes. Concentration of chloride in these samples was determined by a modification of Asper and Schales' titration with $HgNO_3$ (Smith, 1956) for microamounts.

Capacity of the kidney to concentrate urea was studied while the spiny mice were kept on a diet of soybeans without water. Samples were collected daily until the animals lost their appetite or displayed other ill effects. Urea concentration was determined immediately after collection of urine by the microdiffusion method described by Conway (1947).

RESULTS

Body temperatures.—At moderate ambient temperatures both species maintained T_B within the limits typical of mammals, 36 to 38°C (Fig. 1). *A. cahirinus* is capable of doing this from 5 to 30°C, where *A. russatus* demonstrated this ability in the range of 22 to 35°C.

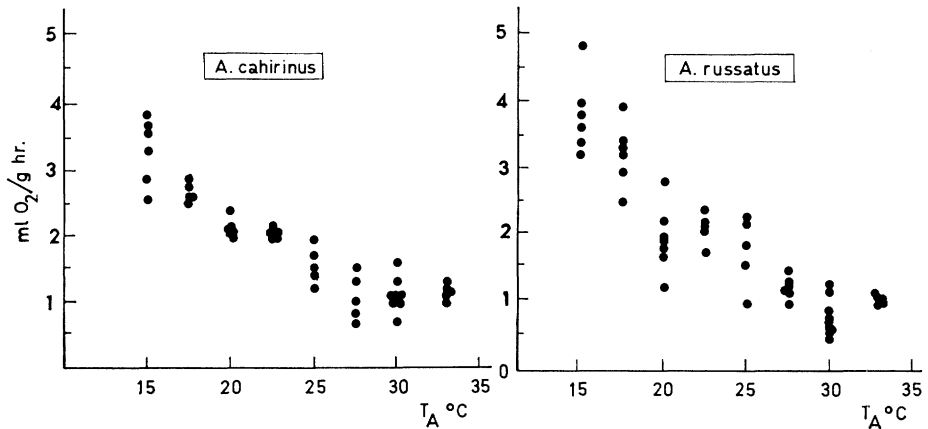


FIG. 2.—Oxygen consumption of spiny mice in relation to ambient temperature. Each point represents one mouse.

In *A. cahirinus*, hyperthermia occurs at $T_A = 32.5^\circ\text{C}$ but *A. russatus* maintained its normal body temperature at T_A up to about 35°C . Both species demonstrated typical behavior at these elevated temperatures—initially extensive activity in search of escape, then calming and lying belly down with limbs stretched out and the body flattened against the substratum as if to increase heat conduction. In both species copious drooling was observed at T_A higher than 34 to 35°C . A T_A of 40°C may be lethal to *A. cahirinus* (Fig. 1). Five out of six individuals of *A. russatus*, however, survived temperature of 42°C for at least 4 hours.

A. cahirinus generally maintains normal body temperature and activity at low ambient temperatures, even during continuous exposure to 5°C , but in *A. russatus* hypothermia is pronounced at low ambient temperatures (Fig. 1). Hypothermic animals may be lethargic. An ambient temperature of 7°C is lethal to *A. russatus* kept singly for more than one night, but they may survive at 5°C if several of them huddle together.

Oxygen consumption.—As demonstrated by data for oxygen consumption at different ambient temperatures (Fig. 2), *A. cahirinus* has a thermoneutral zone beginning at about 27°C and extending to the upper limit of measurements (32.5°C). Minimal metabolic rate was $1.1 \pm \text{SD } 0.2$ ml $\text{O}_2/\text{g hr.}$ In *A. russatus* there was no clear thermoneutral zone; the minimal metabolic rate, measured at 30°C , was $0.80 \pm \text{SD } 0.27$ ml $\text{O}_2/\text{g hr.}$

Effects of water deprivation.—When deprived of drinking water, *A. cahirinus* lost 10 per cent of its body weight in a month at room conditions (18 to 26°C , RH 60 to 75 per cent, 11 to 18 mg $\text{H}_2\text{O}/\text{l air}$). *A. russatus*, however, maintained body weight for a month under identical conditions (Fig. 3). *A. cahirinus* lost 20 per cent of its weight over a period of a month at 26°C and 60 per cent RH (15 mg $\text{H}_2\text{O}/\text{l air}$), whereas *A. russatus* lost about 12 per cent of its weight initially and then stabilized. Neither species could survive without

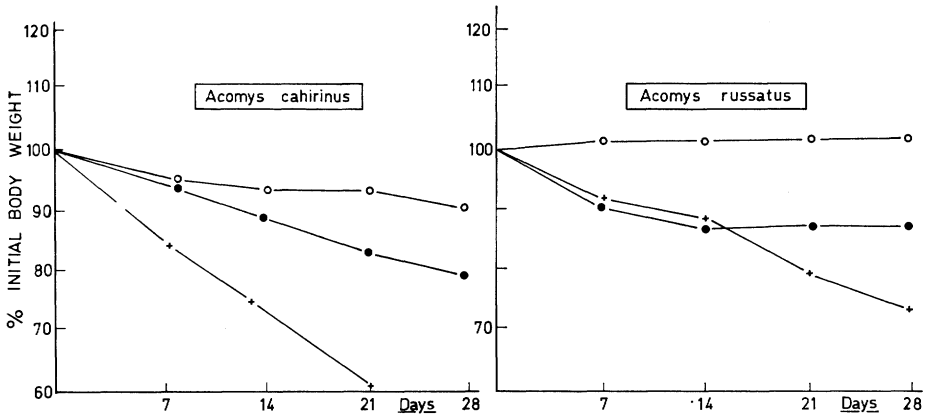


FIG. 3.—Weight changes in spiny mice kept without water on a diet of barley at different temperatures and relative humidities (RH): ○—18 to 24°C, 60 to 75 per cent RH (11–18 mg H₂O/l air); ●—26°C, 50 to 60 per cent RH (15 mg H₂O/l air); + —30°C, 30 per cent RH (10 mg H₂O/l air). Each point represents the mean of seven mice.

water or at 30°C, 30 per cent RH (10 mg H₂O/l air). However, *A. cahirinus* lost weight more rapidly than did *A. russatus* under these conditions.

Mice of both species, after being kept for a month in a negative water balance that caused them to lose 20 to 30 per cent of their initial body weight, rushed to water offered to them. They drank eagerly and continuously for 45 minutes. The six individuals of *A. cahirinus* so tested gained an average amount equal to 11.4 per cent of their weight before drinking. Six *A. russatus* gained 9.3 per cent. After a few days with water *ad libitum*, the weight of individuals of both species was back to the initial level.

Evaporative water loss.—A conspicuous rise in evaporative water loss was obvious in both species at T_A 30°C (Fig. 4). Our methods did not allow us to partition evaporation in the unrestrained animal, but restrained animals at 30°C (with a higher rate of evaporation and a higher oxygen consumption) lost 60 to 70 per cent of their total evaporation through the skin (Table 1). In the white mouse, by comparison, the cutaneous water loss was so small that it could not be detected with the present method (Table 1).

Capacity to utilize NaCl solution and sea water.—Individuals of *A. cahirinus* died when maintained on drinking water of 0.5 N NaCl solution. Those of *A. russatus*, however, survived and maintained body weight indefinitely when fed only soybeans and watered with NaCl solutions as concentrated as 0.8 N. When watered on a NaCl solution of 0.9 N, two of the *A. russatus* tested lost weight, two maintained body weight, and one even gained weight. Several specimens of *A. russatus* were kept on a regimen of soybeans and sea water for more than 6 months with no ill effects (Fig. 5).

Concentrating capacity of the kidney.—The capacity of the spiny mice to concentrate chloride and urea in the urine is shown in Tables 2 and 3. The

TABLE 1.—Cutaneous water loss at 30°C, calculated as the difference between total evaporation while the animal was wrapped in a net bag and in a plastic bag. Data are means of 12 measurements in three different individuals of each species. Data on the white laboratory mouse are included for comparison.

Body wt. (mean g)	Experimental conditions	O ₂ consump. (ml O ₂ /g hr)	Evap. water loss (mg H ₂ O/ml O ₂)	Cut. water loss (mg H ₂ O/ml O ₂)
<i>Acomys cahirinus</i>				
41.3	Control, no bag	1.014	1.962	1.814
	Net bag	1.214	2.699	
	Plastic bag	1.188	0.885	
<i>Acomys russatus</i>				
50.3	Control, no bag	0.835	1.996	1.597
	Net bag	1.328	2.485	
	Plastic bag	1.351	0.888	
<i>Mus musculus</i>				
36.5	Control, no bag	2.208	1.159	—
	Net bag	2.491	1.460	
	Plastic bag	2.510	1.511	

two species differ greatly in their ability to concentrate chloride, but the maximal urea concentration in the urine seems to be similar and amounts to 4700 to 4800 mM urea.

DISCUSSION

A. cahirinus is able to regulate its body temperature over a much greater range of T_A than *A. russatus*. Yet, the superiority of the diurnal *A. russatus* at high ambient temperatures is obvious, both in its capacity to regulate body temperature in a hot environment and its ability to survive extreme temperatures. The upper lethal T_A for *A. russatus* is about 42°C, which is 2°C higher than that for *A. cahirinus* and higher than values recorded for most small rodents (Dawson, 1955; Robinson, 1959; Carpenter, 1966). Another diurnal desert rodent, *Ammospermophilus leucurus*, was found to have a slightly higher lethal ambient temperature (42.6°C) according to Hudson (1962).

A. cahirinus is widely distributed and occurs in rocky habitats throughout Israel (Shkolnik, 1966). It is found not only in deserts, but in high mountains, where air temperatures at night frequently fall below freezing. *A. russatus*, however, is found only in hot deserts. The ability of *A. cahirinus* to regulate its body temperature at lower ambient temperatures as well as its broader thermal neutral zone, when compared with *A. russatus*, is thus correlated with the habitat.

A low metabolic rate often is found in mammals adapted to a hot environment and is common in desert rodents (Dawson, 1955; Murie, 1961; McNab and Morrison, 1963; Carpenter, 1966). A low metabolism would reduce the amount of water required for evaporation and thus be advantageous for life in the desert. Therefore, comparing the data measured in the spiny mice with

TABLE 2.—Urine chloride concentrations found in *Acomys* watered on the highest NaCl concentration on which they could maintain body weight.

Drinking solution	Body weight (per cent of initial weight)	Maximal chloride concentration found in urine (N)
<i>A. russatus</i> (0.9 N)	109	1.480
	100	1.250
	98	1.470
	89	1.390
	87	1.570
<i>A. cahirinus</i> (0.5 or 0.7 N)	105	870
	98	850
	97.5	820
	95.5	650
	92.5	750

the metabolic rate predicted according to well-known "mouse to elephant" curve (Kleiber, 1961) is of interest.

In *A. cahirinus*, the predicted value is 6.1 kcal/day as against 5.4 kcal/day measured, the deviation being 13.5 per cent. In *A. russatus*, the deviation is much more pronounced, 7.2 kcal/day predicted against 4.6 kcal/day measured (34.5 per cent). Such deviation is considerably greater than the 25 per cent reduction reported for the kangaroo rat, *Dipodomys merriami* (Dawson, 1955; Carpenter, 1966).

The failure of spiny mice to maintain their body weight when fed a dry diet in an environment above 25°C was unexpected in view of their wide distribution in extreme desert areas and their abundance in rough, open habitats. In this respect they differ from *Dipodomys* (Schmidt-Nielsen and Schmidt-Nielsen, 1951) and from jirds and gerbils (Fig. 6) living in the deserts of Israel (Shkolnik, 1960).

TABLE 3.—Maximal urea concentrations found in urine of *Acomys* on a diet of dry soybeans. Samples collected after 5 to 7 days of water deprivation.

	Body weight (per cent of initial weight)	Maximal urea concentration (mM)
<i>A. russatus</i>	86.5	3,900
	80.5	4,700
	79.0	4,750
	72.5	4,180
	72.0	4,800
<i>A. cahirinus</i>	85.0	4,000
	82.0	4,170
	78.0	4,700
	73.0	4,080
	71.5	4,700

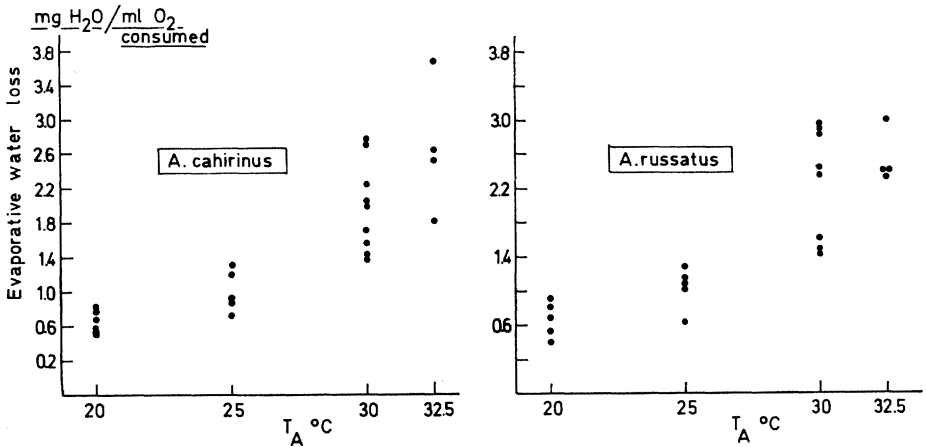


FIG. 4.—Total evaporative water loss of spiny mice in relation to ambient temperature. Each point represents one mouse.

This failure may be explained by their high evaporative water loss. At 25 to 28° they lose approximately two to three times as much water through evaporation as does *Dipodomys merriami* (Schmidt-Nielsen and Schmidt-Nielsen, 1950). Evaporative water loss in spiny mice is high when compared with the highest values recorded for desert rodents at 30°C; 1.5 mg H₂O/ml O₂

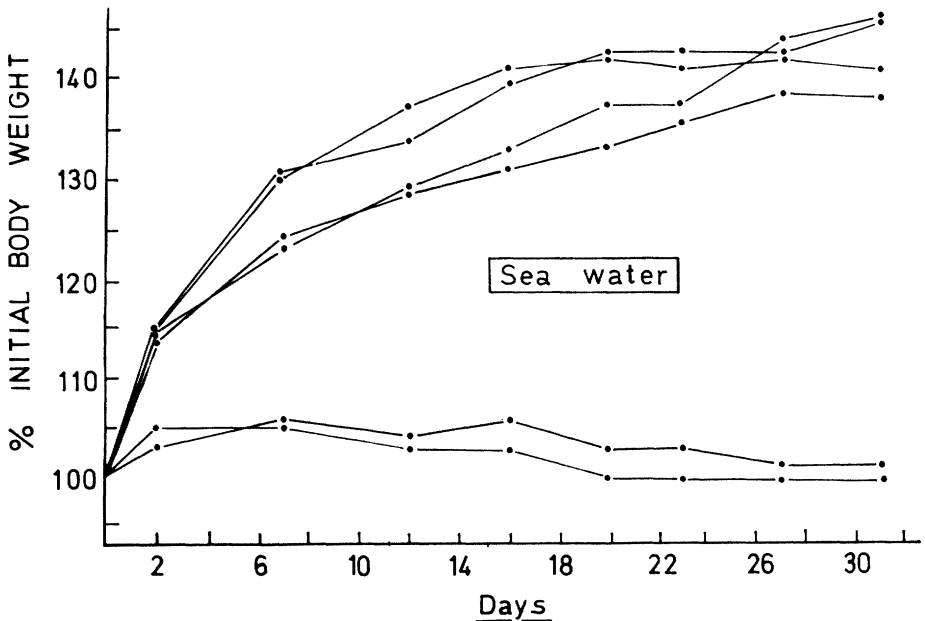


FIG. 5.—Weight changes of *Acomys russatus* while drinking sea water. Each line represents an individual mouse.

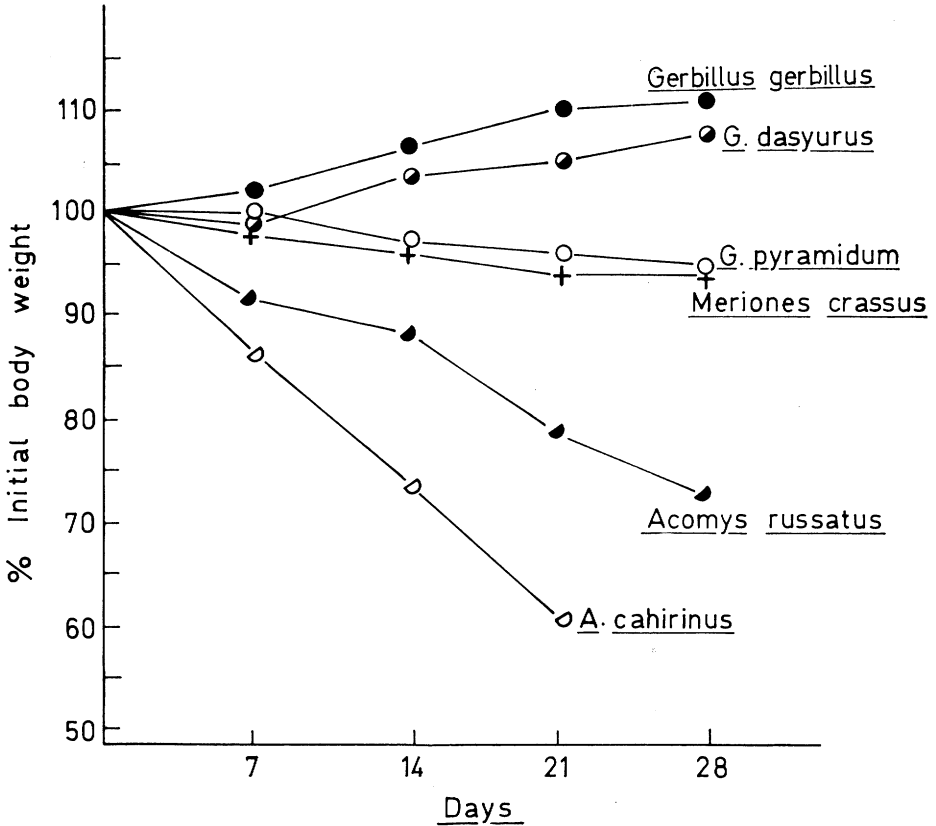


FIG. 6.—Weight changes of various desert rodents living for four weeks on dry barley only, at 30°C and 30 per cent relative humidity. All the mice represented are sympatric with *Acomys* in Israel.

in *Peromyscus eremicus* (MacMillen, 1965), 1.38 mg H₂O/ml O₂ in *Amospermophilus leucurus* (Hudson, 1962). The high rates of evaporation, as compared to other desert rodents, suggest that the ability of *Acomys* to thermoregulate in this fashion may explain their abundance in desert habitats lacking favorable microclimatic conditions. Moreover, it partially solves the physiological problems raised by the diurnal activity of *A. russatus*. In restrained spiny mice, cutaneous evaporation accounted for 60 to 70 per cent of the total. It seems likely that cutaneous water loss also forms a significant part of the total evaporation in unrestrained mice.

In face of the high evaporative water losses the problem of maintaining water balance becomes more acute. It was interesting, therefore, to discover that spiny mice possess a kidney with an outstandingly efficient mechanism for urine concentration. Urea concentrations in the urine of spiny mice (up to 4700 to 4800 mM) are among the very highest known in mammals (Schmidt-Nielsen and O'Dell, 1961; MacMillen and Lee, 1967). Considering

the fact that land snails are one of the main components of the diet of spiny mice in nature (Shkolnik, 1966), the exceptional capacity of the kidney to concentrate urea renders this food component a source for a relatively abundant supply of water.

Assuming that total electrolytes in urine roughly equals twice the concentration of Cl^- , the kidney of *A. russatus* is capable of concentrating electrolytes to about 3000 miliosmols, which is also an exceptionally high concentration among mammals. In this respect, *A. cahirinus* is far less capable and its maximal ability to concentrate electrolytes to 1700 milliosmols is common among other desert rodents, for example, *J. jaculus*, *G. gerbillus*, and *Psammomys obesus* (Schmidt-Nielsen and O'Dell, 1961).

The ability of *A. russatus* to excrete concentrated solutions of electrolytes enables this animal to derive water from the sap of succulent desert plants that often have a high electrolyte content. *A. russatus* has often been observed eating such succulent plants (*Atriplex halimus*, *Haloxylon articulatum*, *Anabasis articulatum*). The ability of *A. russatus* to derive water from concentrated solutions of electrolytes was clearly shown by its ability to survive on drinking water of NaCl solutions of 0.7 to 0.9 N.

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